Metamaterial Absorber for X-band application

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***Abstract*—In this work, a wideband absorber structure is presented. The proposed structure is composed of an outer ring resonator along with a diamond shaped patch structure. The structure exhibits an absorption greater than 93% over the frequency range of 7.87 GHz to 10.10 GHz.**

I. Objective

The objective of this work is to design a single- layer low profile wide-band metamaterial absorber having the following:

1) Operating Frequency of 8-10 GHz
2) Absorption $\geq 90\%$ in the operating frequency band
3) Thickness $\leq \lambda/10$

II. Methodology

Electromagnetic metamaterials are three-dimensional, artificially engineered periodic structures composed of sub-wavelength "unit cell" that exhibit effective constitutive parameters—such as permittivity and permeability—not typically found in nature. Whereas, a metasurface is the two-dimensional version of the metamaterial. It consists of a single or fewer layer of sub-wavelength elements arranged on the surface. By engineering the geometric architecture of the unit cell, the material's effective constitutive parameters can be tailored to precisely manipulate the complex transmission (S_{21}) and reflection (S_{11}) coefficients. Based on these scattering parameters, the absorptivity of the metasurface is analytically determined by

$$Abs = 1 - |S_{11}|^2 - |S_{21}|^2 \tag{1}$$

Where Abs is the absorbed power, $|S_{11}|^2$ is the reflected power, $|S_{21}|^2$ is the transmitted power. Since the proposed metasurface is backed by a metal plate the transmission coefficient is negligible ($S_{21} \approx 0$). so, We need $|S_{11}| \leq \sqrt{0.1} \approx 0.316$. The reflection coefficient is also depend on the input impedance Z_{in} of the metasurface as

$$S_{11} = \frac{Z_{in} - \eta}{Z_{in} + \eta} \tag{2}$$

Whereas, η is the free space impedance, 377Ω. It is computed that to achived more than 90% absorption as per the objective, the input impedance (Z_{in}) should be,

$$196\Omega \leq Re(Z_{in}) \leq 725\Omega \tag{3}$$

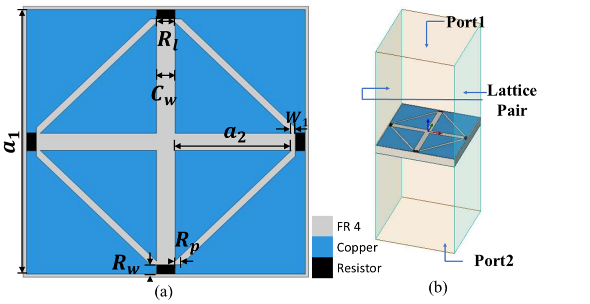

Fig. 1. The schematic of the proposed unit cell in (a) top view and (b) trimetric view of simulate setup.The optimized geometrical parameters are $a_1 = 14.65$ mm, $a_2 = 6.145$ mm, $R_l = 1$ mm, $R_w = 0.5$ mm, $R_p = 0.3$ mm, $W_1 = 0.2$ mm, and $C_w = 0.96$ mm.

A. *Design analysis*

The schematic of the unit cell of the proposed absorber is shown in Fig. 1(a). The metallic pattern of the unit cell consists of an outer ring resonator along with a diamond shaped patch structure. Copper (conductivity $\sigma = 5/8 \times 10^7$ S/m) is used for the metal layers. The dielectric spacer between them is FR4, with a height of 1.6 mm relative permittivity (ε_r) of 4.4 and a loss tangent ($\tan\delta$) of 0.02. It should be mentioned that the substrate thickness is 1.6 mm, corresponding to approximately $\lambda_0/21$ at 9 GHz. The unit cell is integrated with four lumped resistors, 300Ω as shown in Fig. 1(a). The metasurface is backed by a continuous metallic ground plane. To comprehend the absorbtion performance of the proposed structure, the numerical simulation of the proposed design is executed using Ansys HFSS. In the simulations, as shown in Fig. 1(b), we have utilized Floquet ports to excite the structure accompanied by lattice pair boundary conditions for realizing an infinite array condition. The optimized geometrical parameter are enlisted in the caption of Fig. 1(a). Since the designed absorber has fourfold symmetry, it exhibits similar electromagnetic behavior for both TE- and TM-polarized incident waves.

The reflection coefficient and the real part of the input impedance of the absorber are shown in Fig. 2(a). It is observed that the real part of the input impedance also falls within the aforementioned calculated range mentioned in Eq(3). As a result, the reflection coefficient $S_{11}(dB)$ is also less than 10 dB throughout the operating region from 8 to 10 GHz. As the metasurface is backed by the solid metal plane, the transmission coefficient S_{21} is zero. Thus, from the equation (1), the absorption can be expressed as $Abs = 1 - |S_{11}|^2$. The absorption of the proposed metasurface is exhibited in Fig. 2(b). It is noticed from the Fig. 2(b) that in our operating band, i.e, 8 to 10 GHz, the absorption is more

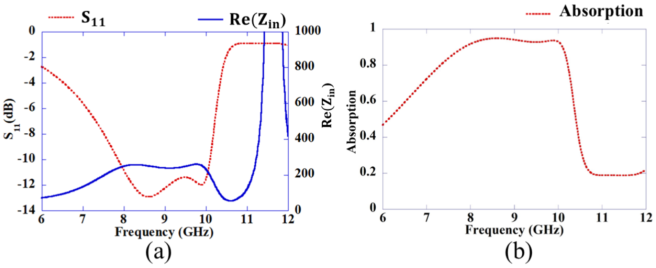

Fig. 2. (a) Reflection Coefficient and Real part of the input impedance and (b) Absorption property of the proposed metasurface

than 90% as per the requirement.

B. Design Stage Analysis

In order to understand the chronological development of the metasurface, a step-by-step analysis has been performed. Initially, a square ring-shaped resonator backed by the continuous metal plane, as shown in 'Stage 1' in Fig. 3(a), is analyzed. It is observed that the structure of 'Stage 1' is resonated at around 10.5 GHz with very less resonant strength, as shown in Fig. 3(b). Further investigate this S_{11} result, it is observed that the real part of input impedance is around 10Ω, as illustrated in Fig. 3(c). To enhance the input impedance, a triangular structure is integrated with the outer square ring, as indicated as 'Stage 2' in Fig. 3(a). The triangular structure increases ohmic losses, thereby enhancing the resistivity of the structure as exhibited in 3(c). As a result, it is observed that the unit cell resonated at around 10.5 GHz. However, it also noticed that at the desire frequency range, the real part of the impedance is much lower than the required value expressed in Eq (3). To enhance the resistivity of the structure, four lumped resistance 300Ω are integrated with the unit cell as indicated as stage 3 Fig. 3(a). By incorporating this lumped resistor, the real part of the input impedance increases. As a result, the unit cell of the stage 3 resonate at around 8.5 GHz. However, to achieve the wide absorption bandwidth, another resonator should be incorporated in the structure. In this regard, resonator indicated as stage 4 in Fig. 3(a) has been analyzed. It is observed in Fig. 3(c) that the structure resonated at around 9.5 GHz. As a final stage, we integrate the resonator of stage 3 and stage 4 as shown in the Fig. 1. As a result of this incorporation, more than 90% absorption achieved between 8 to 10 GHz frequency

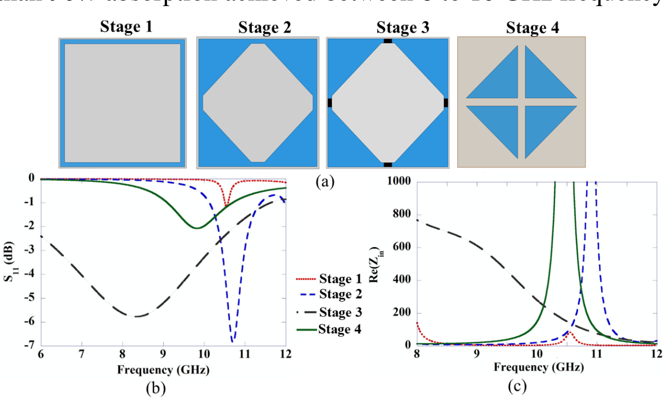

Fig. 3. The chronological development of the absorber: (a) Schematic view, (b) Magnitude of the reflection coefficient and (c) Real part of the input impedance of each stage.

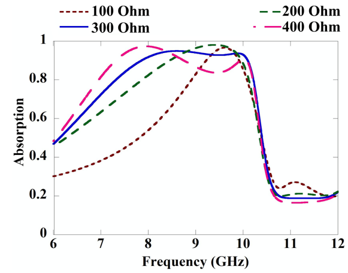

Fig. 4. Effect of variation of lumped resistance over the absorption property of the proposed metasurface.

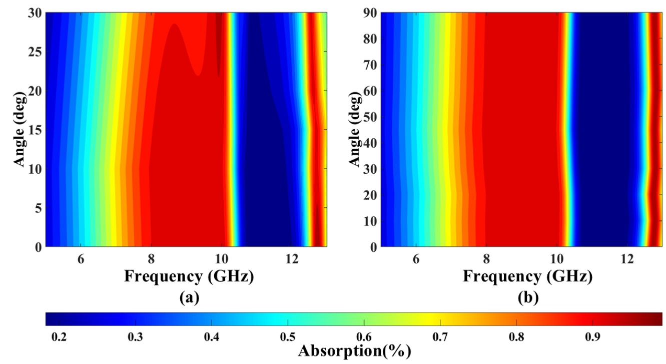

Fig. 5. Absorption property of the proposed metasurface with (a) different oblique incident angle and (b) different polarization angle.

exhibitted in Fig. 2(b). The absorber exhibits an absorption greater than 90% over the frequency range of 7.87–10.10 GHz.The absorption response exhibits a local minimum within the operating band, with a minimum value of 93%.

In order to understand the effect of the lumped resistance in the absorption characteristics of the proposed absorber, a parametric analysis has been performed and exhibited in Fig. 4. It is observed that the lumped impedance of 300Ω offers a steady absorption curve over 93% absorbance throughout the frequency region. The incident angle stability of the absorber is important from the perspective of practical applications. Simulated absorbance for different incident angles from 0^o to 30^o is exhibited in Fig. 5(a). It can be observed from Fig. 5(a) that the absorber provides a stable performance fupto 20^0. Apart from that, the polarization sensitivity is also studied and exhibited in Fig. 5(b). Due to the four fold symmetry of the structure, it offers a polarization insensitive behavior.

III. Conclusion

In this work, a single-layer, low-profile absorber for X-band operation is presented. The proposed design achieves effective impedance matching, resulting in absorption above 90% from 7.87 to 10.10 GHz with a minimum absorption of 93% and an electrical thickness of approximately $\lambda_0/21$. Owing to its fourfold symmetry, the absorber exhibits polarization insensitive behavior and maintains stable absorption for oblique incidence up to $\pm 20^o$ making it suitable for practical X-band applications.